



Foundations of **Platform-Assisted Auctions**

Elaine Shi
Carnegie Mellon University

Joint work with Hao Chung and Ke Wu

Ad space on websites



"Best Air Purifier for Your Money According to Doctors"

-Money Magazine



AdChoices

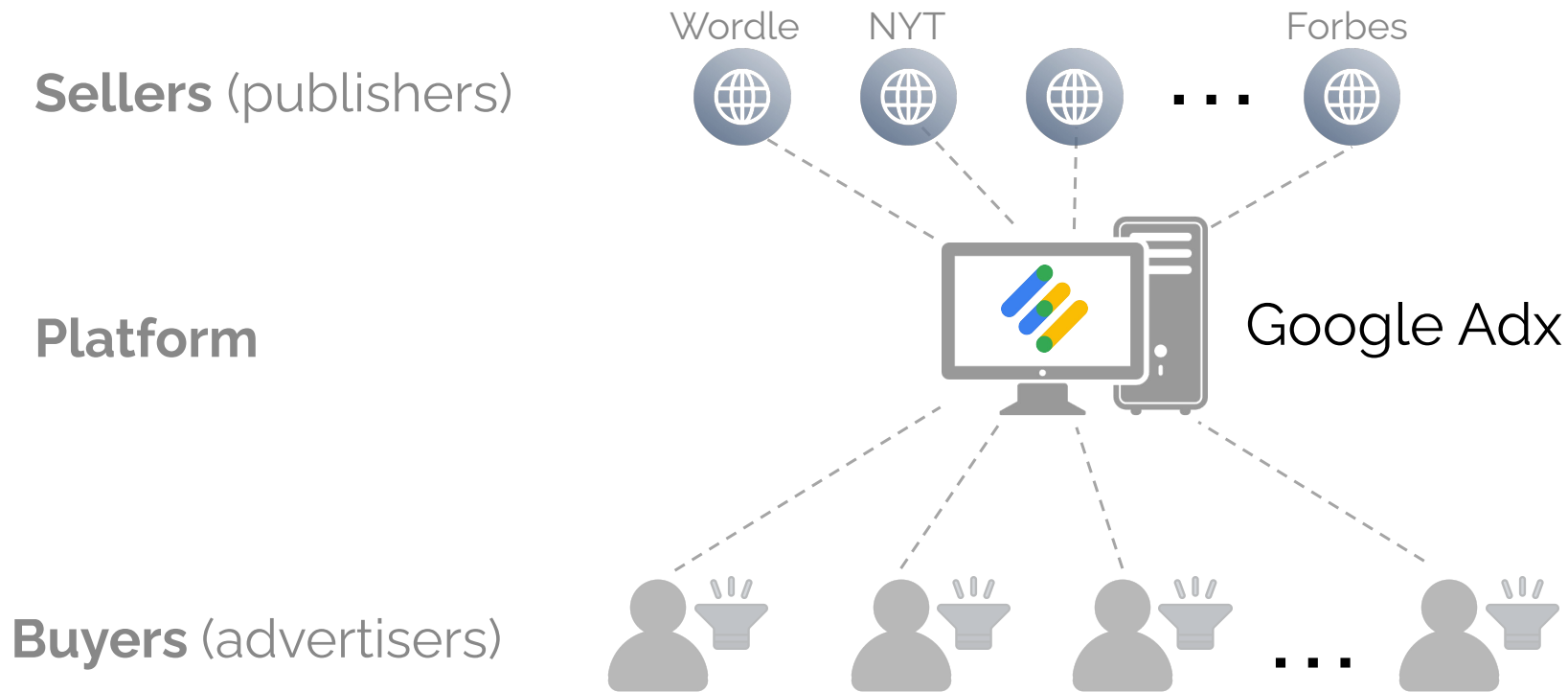


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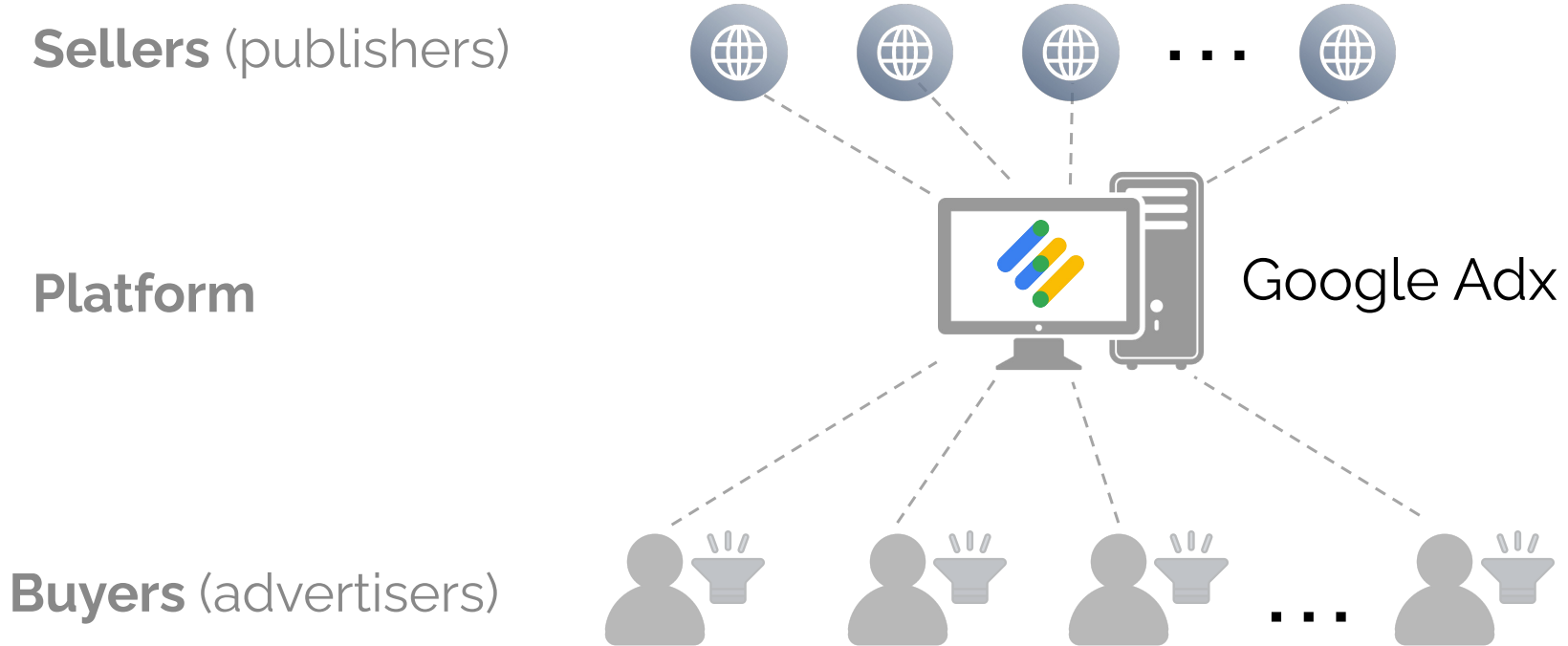
M	E	D	I	A
T	E	S	L	A

Q W E R T Y U I O P

Platform-assisted auctions



Platform gets **remunerated** for **value-added services**:
→ rendezvous, search, recommender system, payment processing



Justice Department Sues Google for Monopolizing Digital Advertising Technologies

Google accused of

- **withholding** seller revenue
- **injecting bids** to raise price

.....

<https://www.justice.gov/archives/opa/press-release/file/1563746/dl>

New theory of **anti-trust** auction design

New theory of anti-trust auction design



Auction literature

- Trusted auctioneer
- Assume no collusion
- Permissioned

New theory of anti-trust auction design



Auction literature

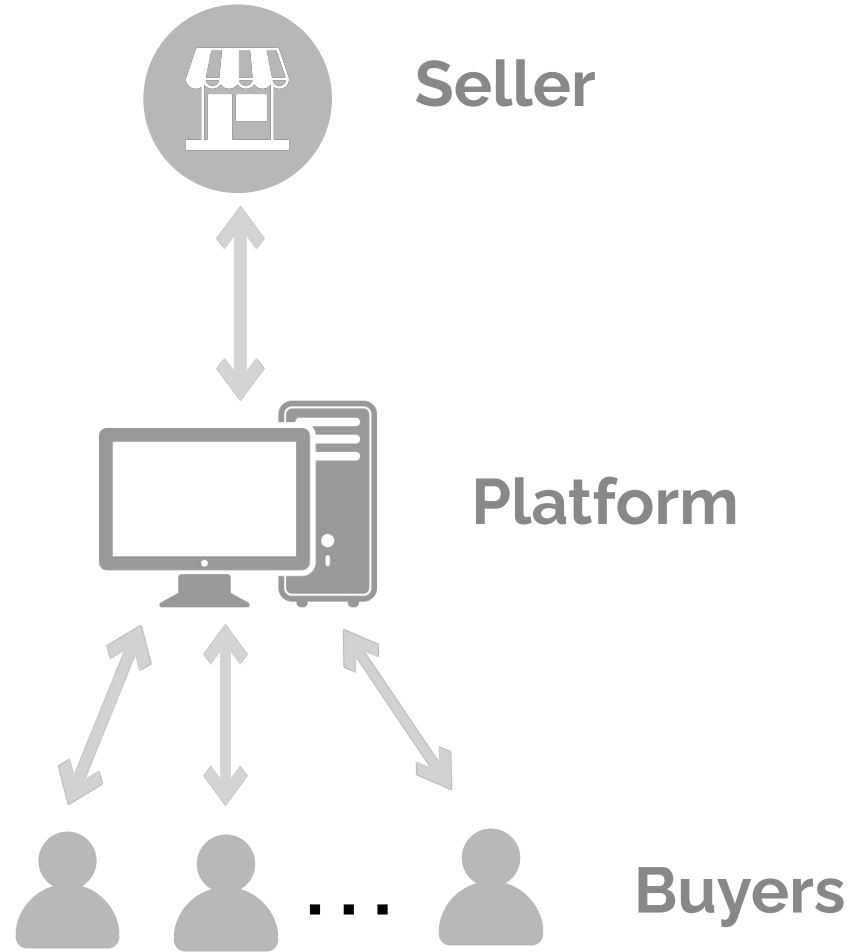
- Trusted auctioneer
- Assume no collusion
- Permissioned



Reality

- Trustless environment
- Collusion made easy
- Permissionless

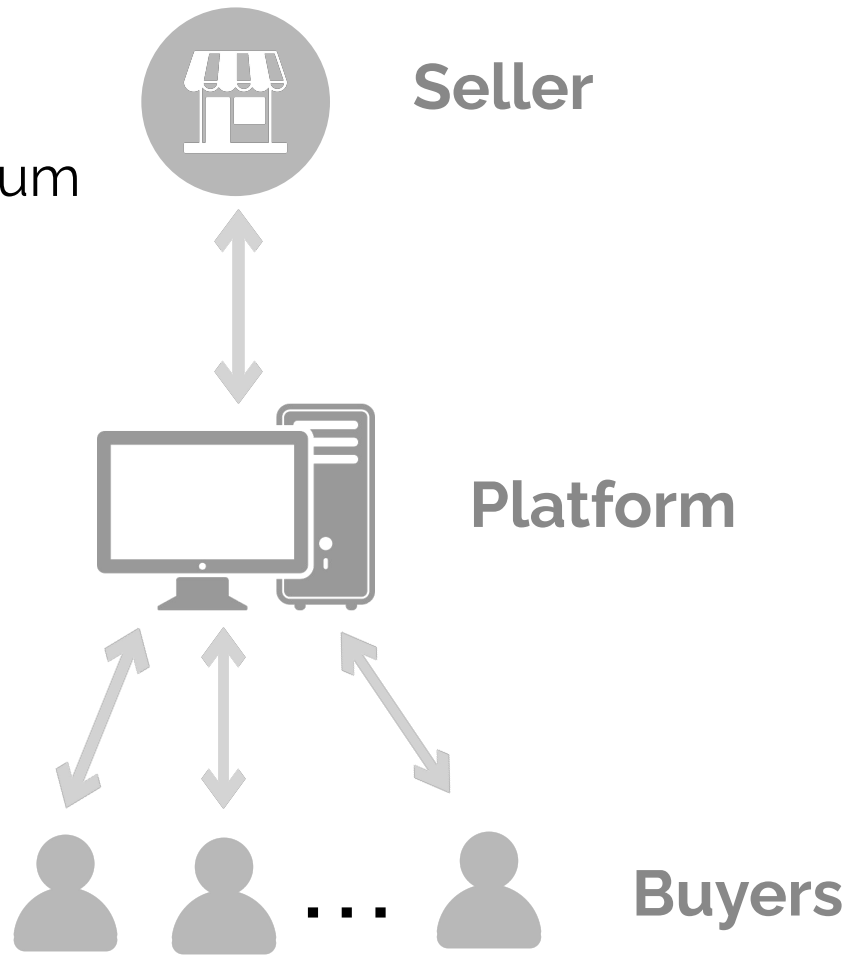
What is a **dream**
platform-assisted auction?



Incentive compatible (IC)

→ **Honest** is best response/equilibrium

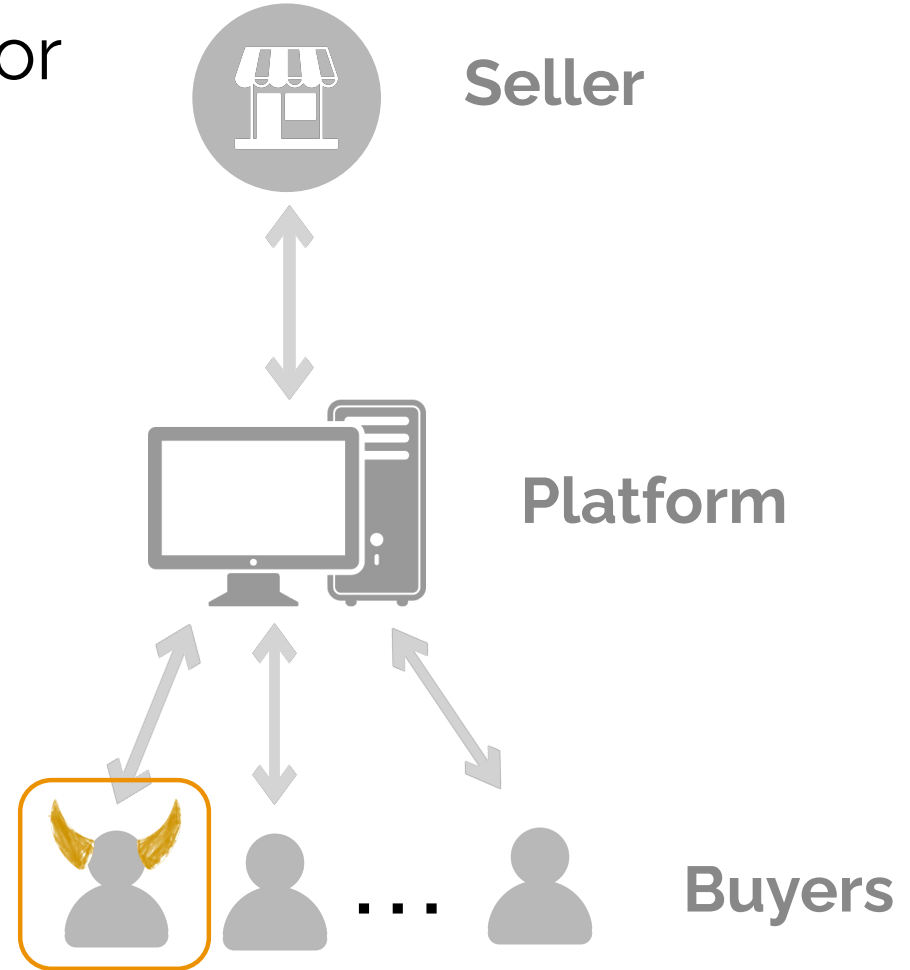
What is a **dream**
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Incentive compatible (IC) for

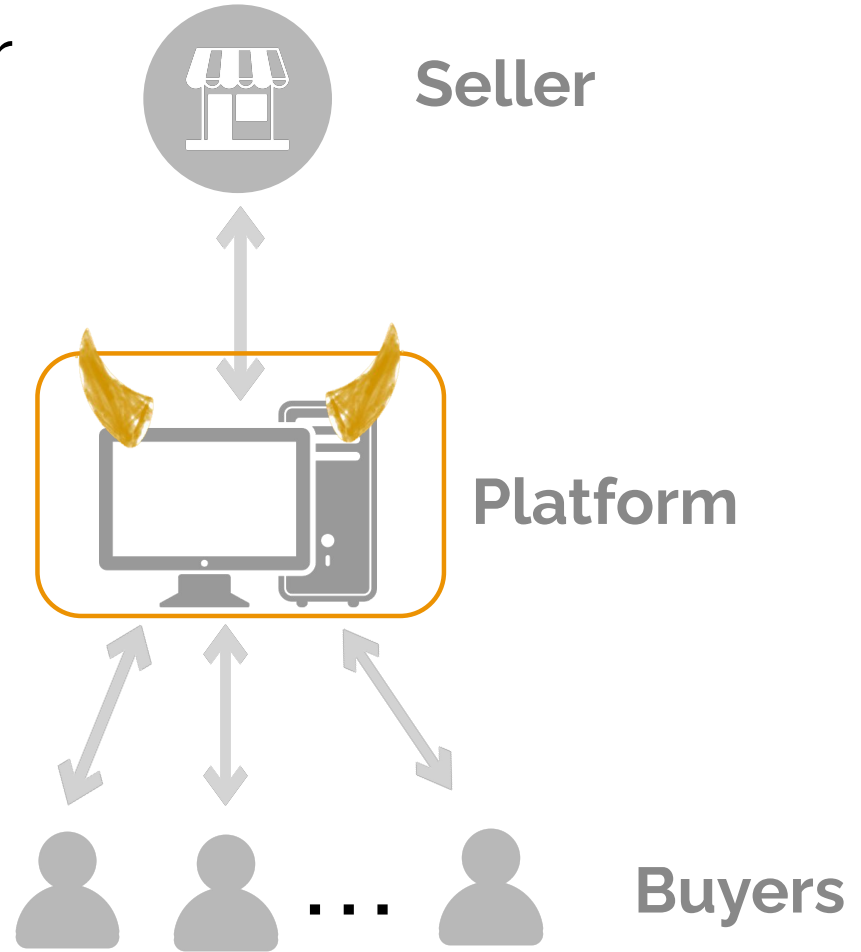
- a buyer

What is a **dream**
platform-assisted auction?



Incentive compatible (IC) for

- a buyer
- the platform

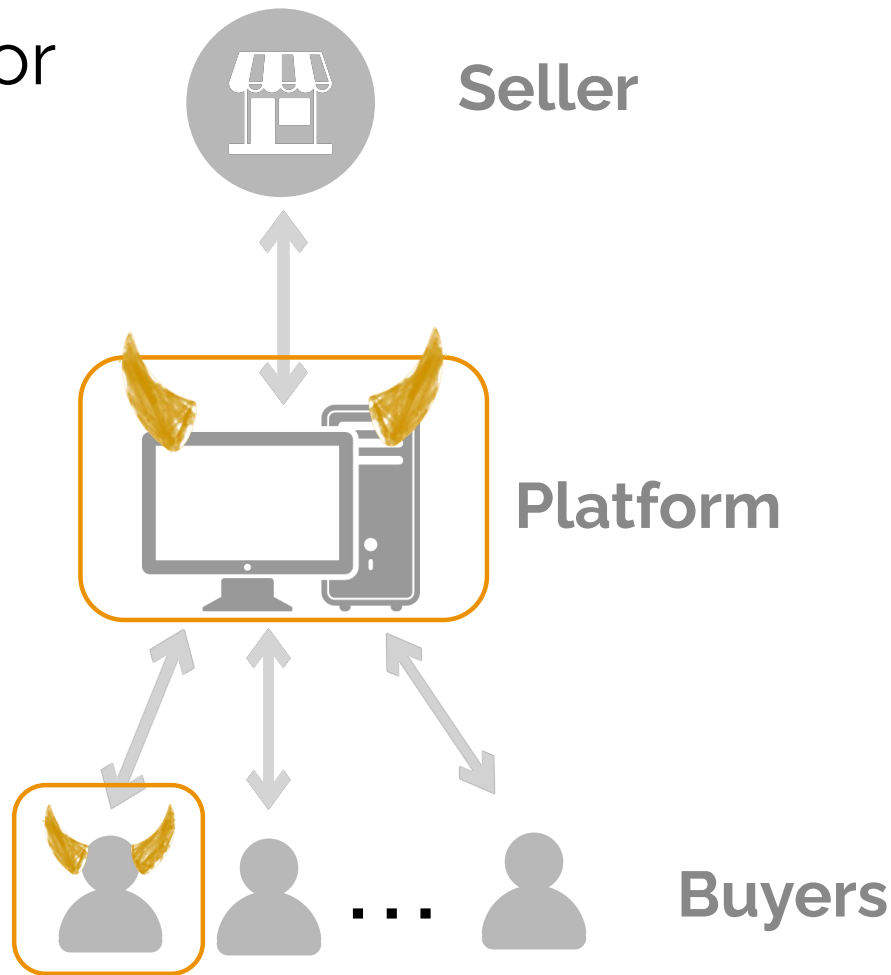


What is a **dream**
platform-assisted auction?

Incentive compatible (IC) for

- a buyer
- the platform
- platform-buyer coalition

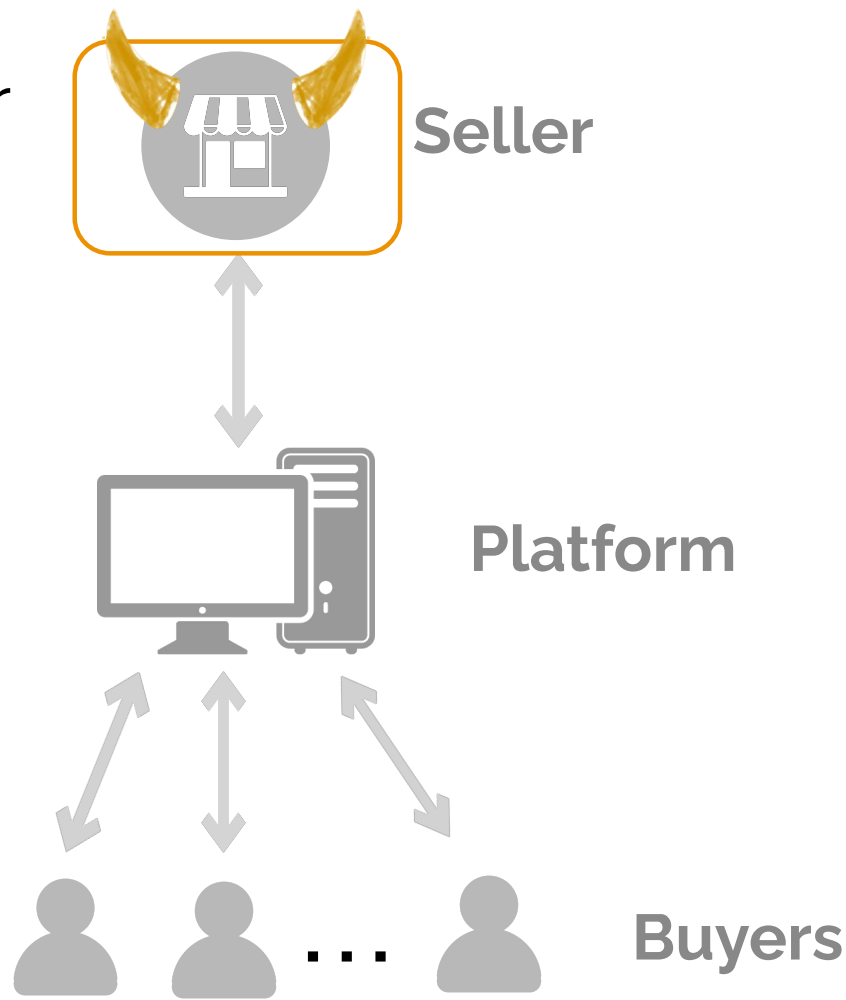
What is a **dream**
platform-assisted auction?



Incentive compatible (IC) for

- a buyer
- the platform
- platform-buyer coalition
- the seller

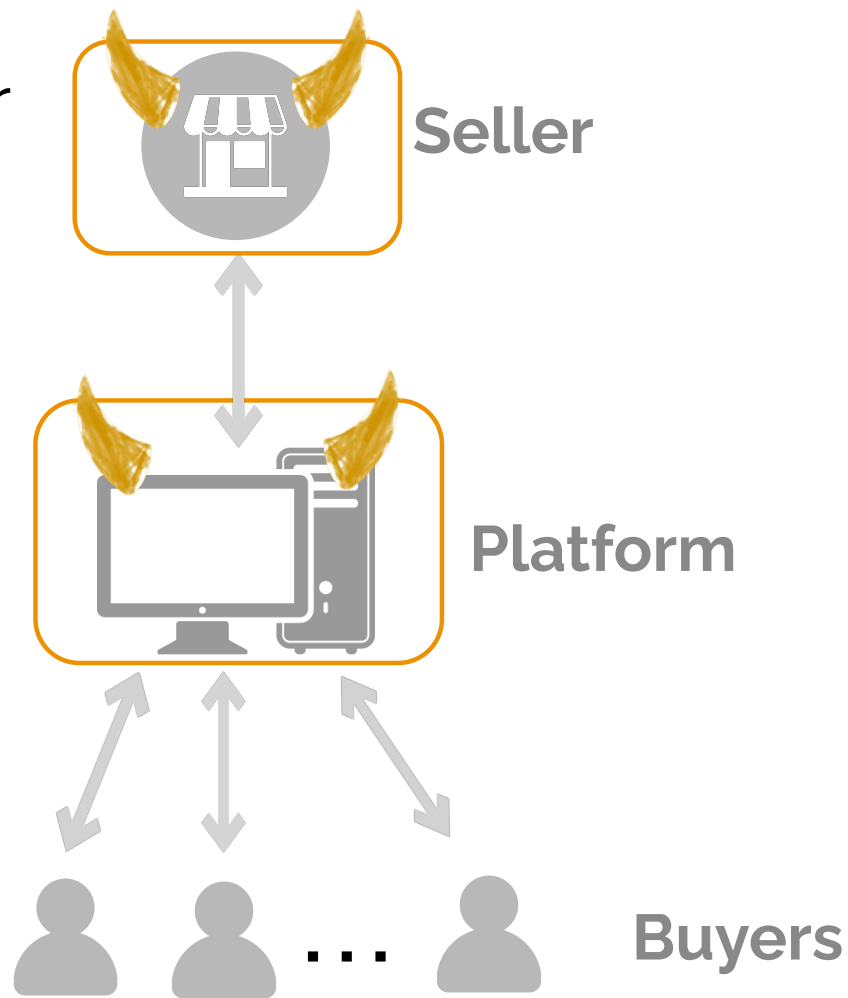
What is a **dream**
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Incentive compatible (IC) for

- a buyer
- the platform
- platform-buyer coalition
- the seller
- platform-seller coalition

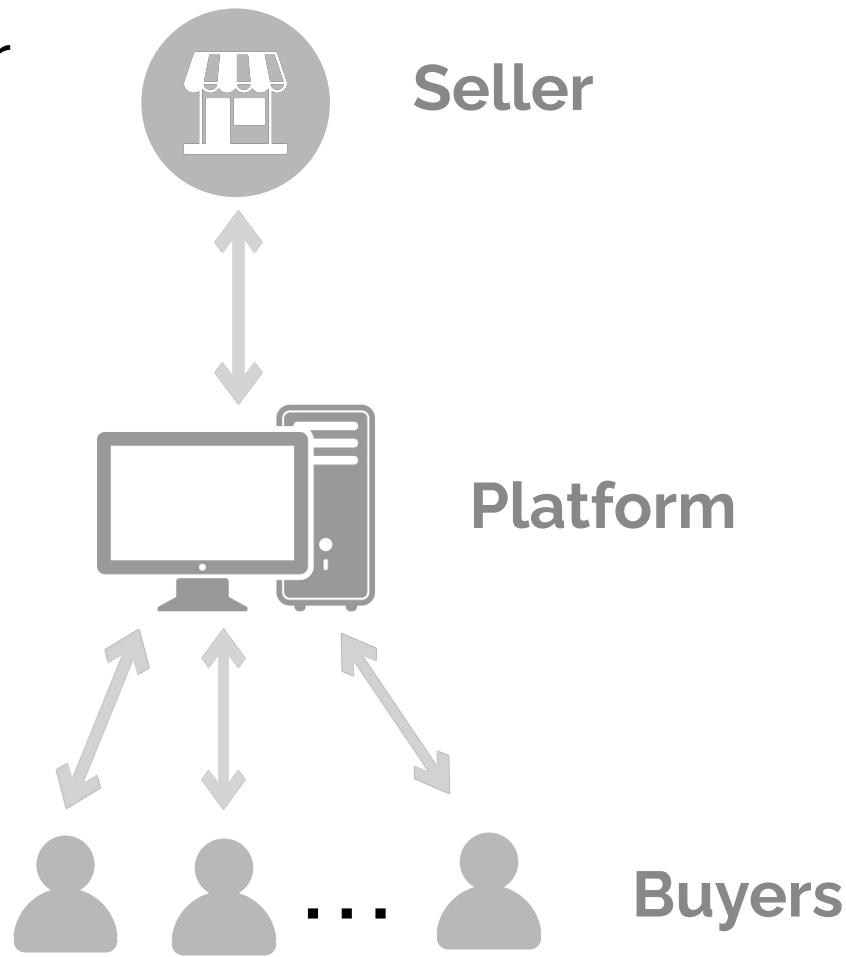
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Incentive compatible (IC) for

- a buyer
- the platform
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- the seller
- platform-seller coalition

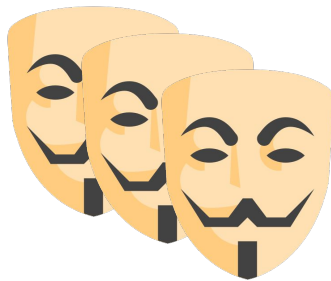
What is a **dream**
platform-assisted auction?



Overbid,
underbid



Fake bids



Arbitrarily
deviate from
protocol

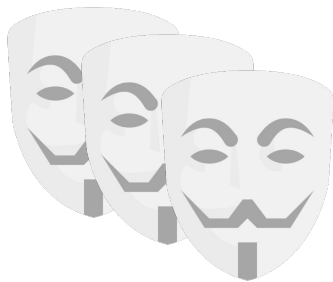


Strategy space

Overbid,
underbid



Fake bids



Arbitrarily
deviate from
protocol



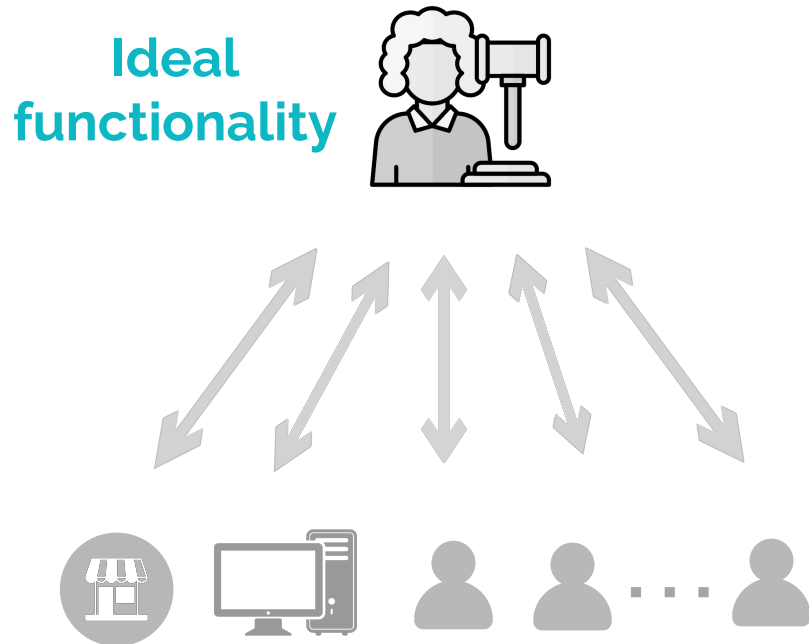
Assumption:



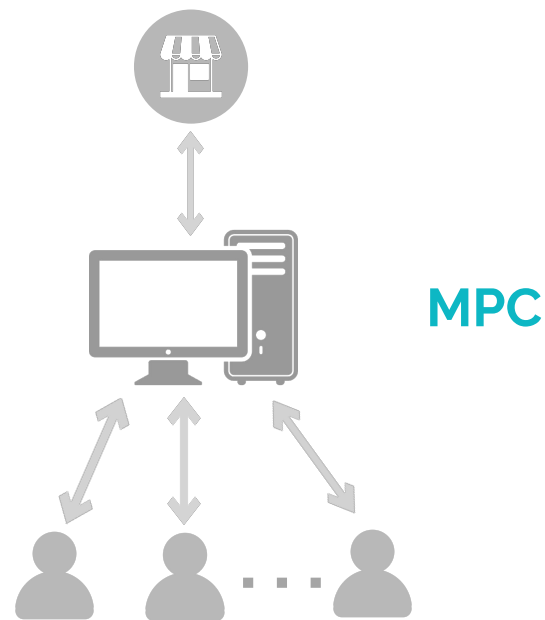
- ✓ cares about reputation
- ✓ adopts only **safe** strategies that do not risk detection

Why not just use MPC?

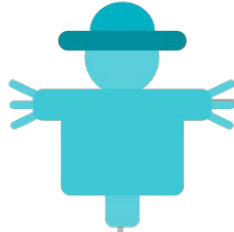
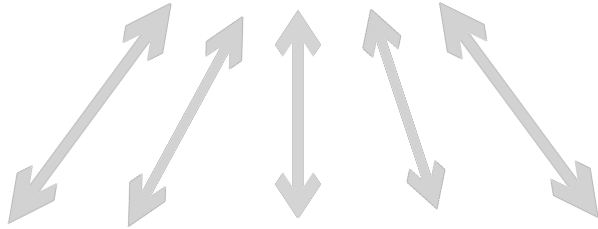
Ideal world



Real world

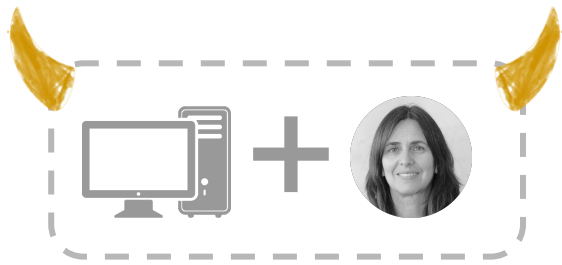


Ideal world

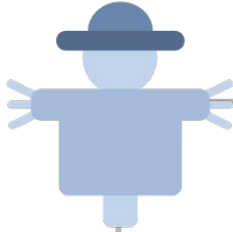


2nd price auction

- Allocate to **top** bidder
- Each pays **2nd** price
- Platform gets **10%** of revenue, seller gets the rest

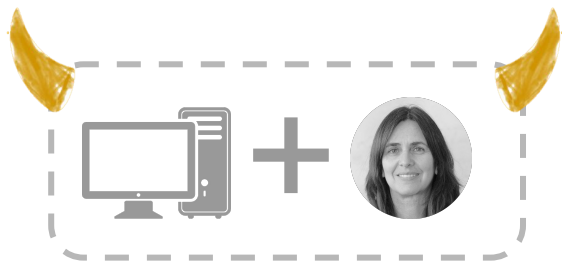


benefit from overbidding

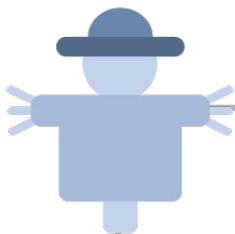


2nd price auction

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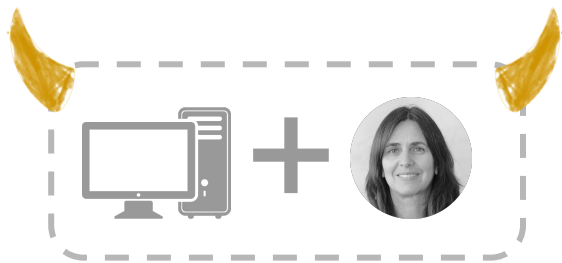


Example: 2 buyers



2nd price auction

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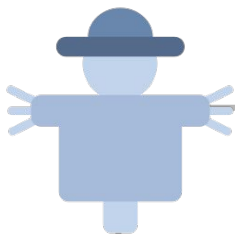
Example: 2 buyers



value = 5

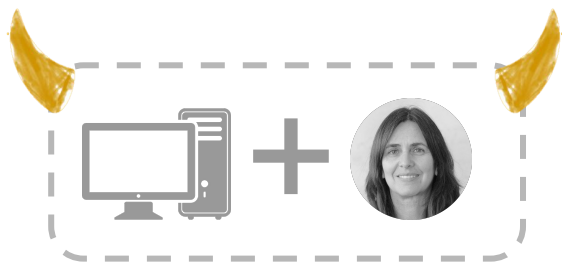


value = 8



2nd price auction

- Allocate to top bidder
- Each pays 2nd price
- Platform gets 10% of revenue, seller gets the rest



Example: 2 buyers



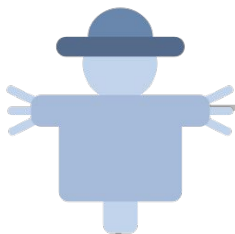
value = 5



value = 8

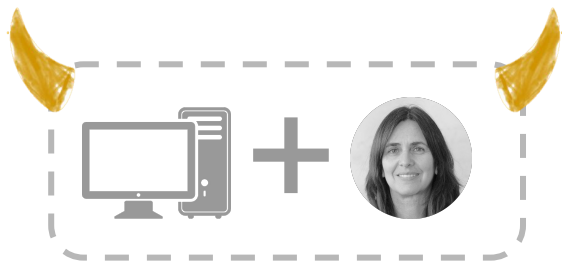


should bid **8 - ϵ**



2nd price auction

- Allocate to top bidder
- Each pays 2nd price
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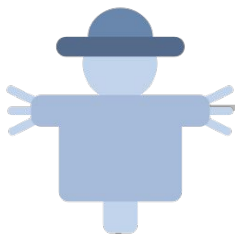
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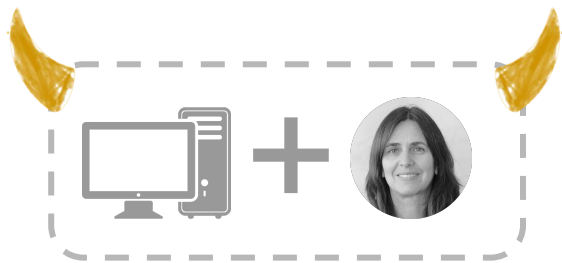


value $\xleftarrow{\$}$ [0, 10]



2nd price auction

- Allocate to top bidder
- Each pays 2nd price
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Example: 2 buyers



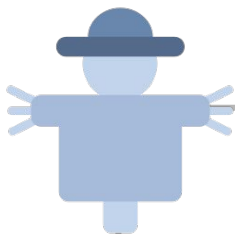
value = 5



value $\leftarrow^{\$}$ [0, 10]



should bid **5.45**



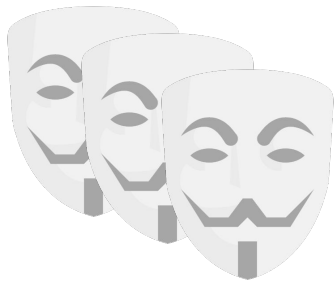
2nd price auction

- Allocate to top bidder
- Each pays 2nd price
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Overbid,
underbid



Fake bids



Arbitrarily
deviate from
protocol



**No
protection**

MPC dos and don'ts

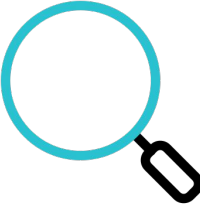
Can we have a **dream** platform-assisted auction?

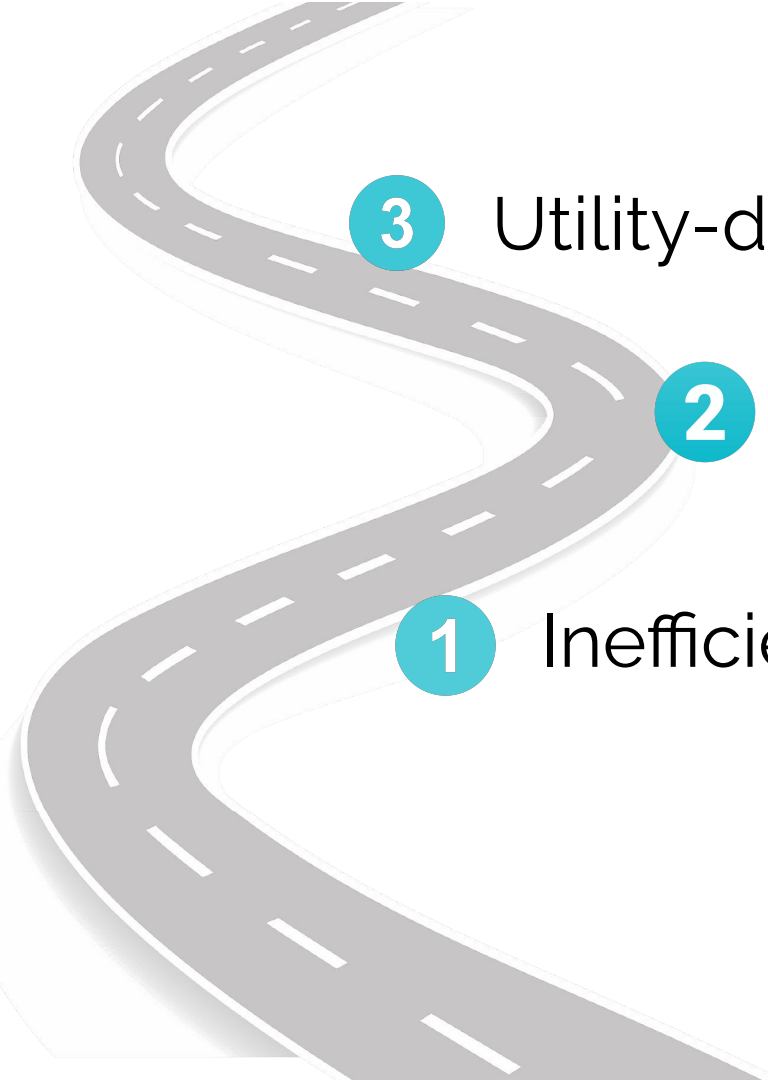
Crypto



**Mechanism
design**

“**Decentralized** mechanism design”

 Our Results



3

Utility-dominated emulation

2

Fundamental limitations

1

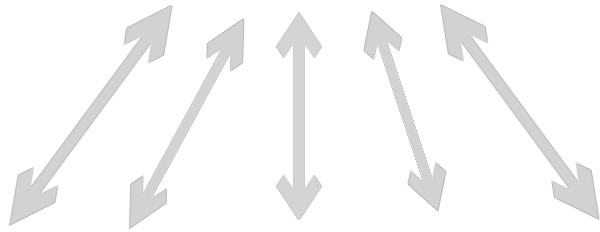
Inefficient MPC-based auction



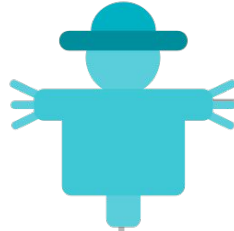
Our Results

Recall the strawman MPC protocol

Ideal world



...

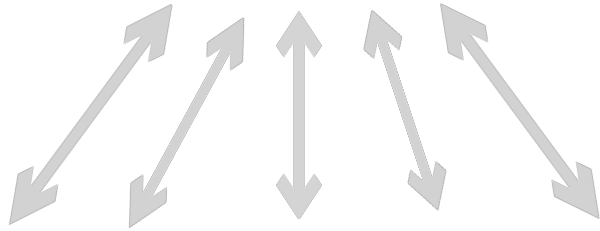


2nd price auction

- Allocate to top k bidders
- Sale price = $(k+1)$ -st price
- Platform gets 10%
- Everyone learns their private outcome

The fix

Ideal world



...



2nd price with **reserve R**

- Allocate to top k bidders **who bid $\geq R$**
- Sale price = **$(k+1)$ -st price or R , whichever greater**
- Platform gets **nothing**
- **Broadcast final price** to all

1

2

3

IC for:

✓ buyer

2nd price with reserve R

- Allocate to top k bidders **who bid $\geq R$**
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IC for:

✓ buyer ✓ platform

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IC for:

✓ buyer ✓ platform

✓ platform-buyer coalition

2nd price with reserve R

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IC for:

✓ buyer ✓ platform

✓ platform-buyer coalition

What can the platform
do that the buyer
cannot on its own?

2nd price with reserve R

- Allocate to top k bidders
who bid $\geq R$
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 R , whichever greater
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Broadcast prevents the “partitioned world” attack



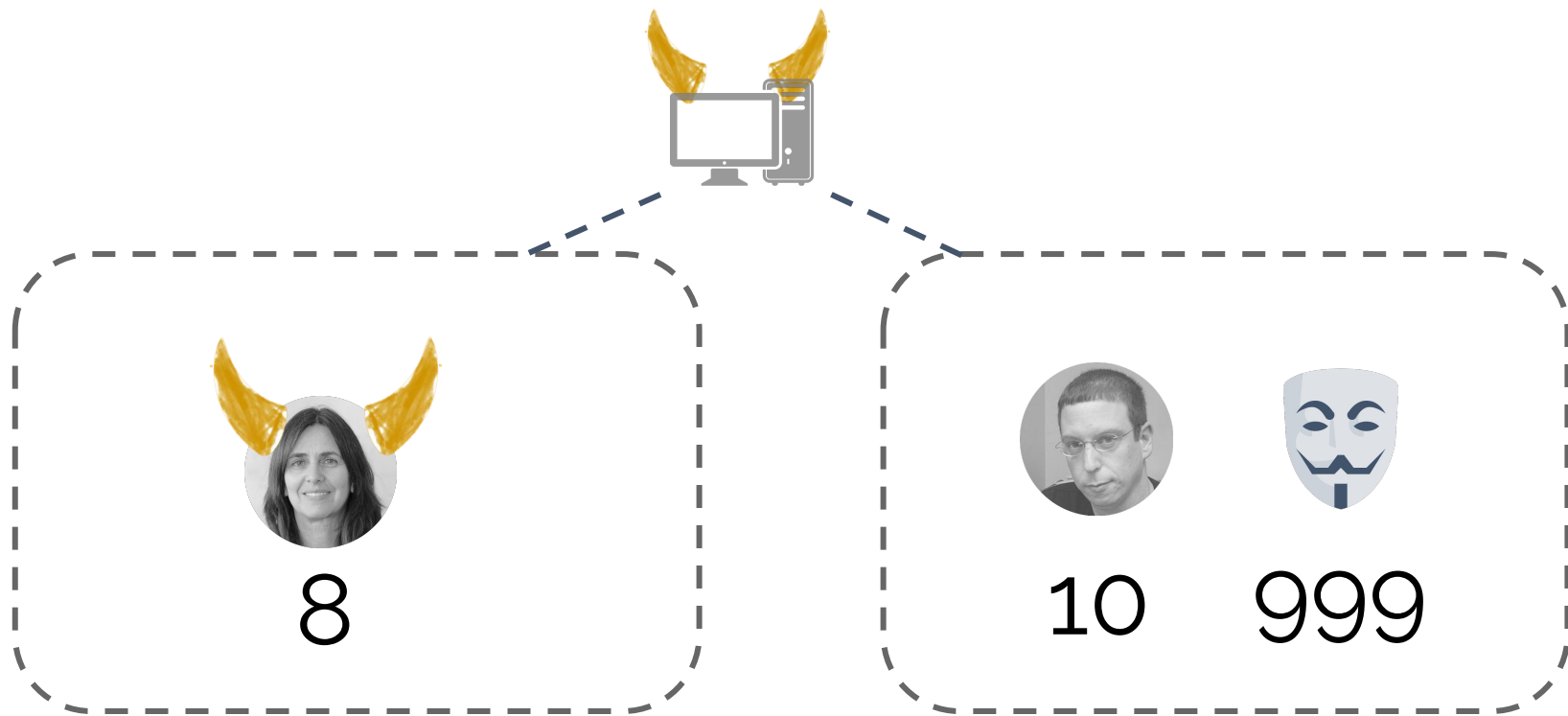
8



10

1 item, reserve = 0

Broadcast prevents the “partitioned world” attack



1 item, reserve = 0

IC for:

- ✓ buyer ✓ platform
- ✓ platform-buyer coalition

Bayesian IC for:

- ✓ seller
- ✓ platform-seller coalition

assume: **suitable reserve**

2nd price with reserve R

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IC for:

- ✓ buyer ✓ platform
- ✓ platform-buyer coalition

Bayesian IC for:

- ✓ seller
- ✓ platform-seller coalition



Bake optimal price floor
into mechanism itself

assume: **suitable reserve**

2nd price with reserve R

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IC for:

- ✓ buyer ✓ platform
- ✓ platform-buyer coalition

Bayesian IC for:

- ✓ seller
- ✓ platform-seller coalition

Revenue optimal!

assume: **suitable reserve**

2nd price with reserve R

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- **Broadcast final price** to all

Privacy of MPC is important!

Bayesian IC for:

- ✓ seller
- ✓ platform-seller coalition

Revenue optimal!

assume: **suitable reserve**

2nd price with **reserve R**

- Allocate to top k bidders **who bid $\geq R$**
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IC for:

- ✓ buyer ✓ platform
- ✓ platform-buyer coalition

Bayesian IC for:

- ✓ seller
- ✓ platform-seller coalition

Revenue optimal!

assume: **suitable reserve**

Summary

2nd price with reserve R

- Allocate to top k bidders **who bid $\geq R$**
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Limitations

2nd price with reserve R

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Different fee
structure



Limitations

2nd price with reserve R

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Limitations

Different fee
structure



Avoid the
broadcast



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Limitations

Different fee
structure



Avoid the
broadcast



Improve
efficiency



2nd price with reserve R

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Impossible

Different fee structure



Avoid the broadcast



Improve efficiency

2nd price with reserve R

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Utility dominated emulation

Different fee structure



Avoid the broadcast



Improve efficiency



2nd price with reserve R

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Generic MPC incurs n^2 cost!

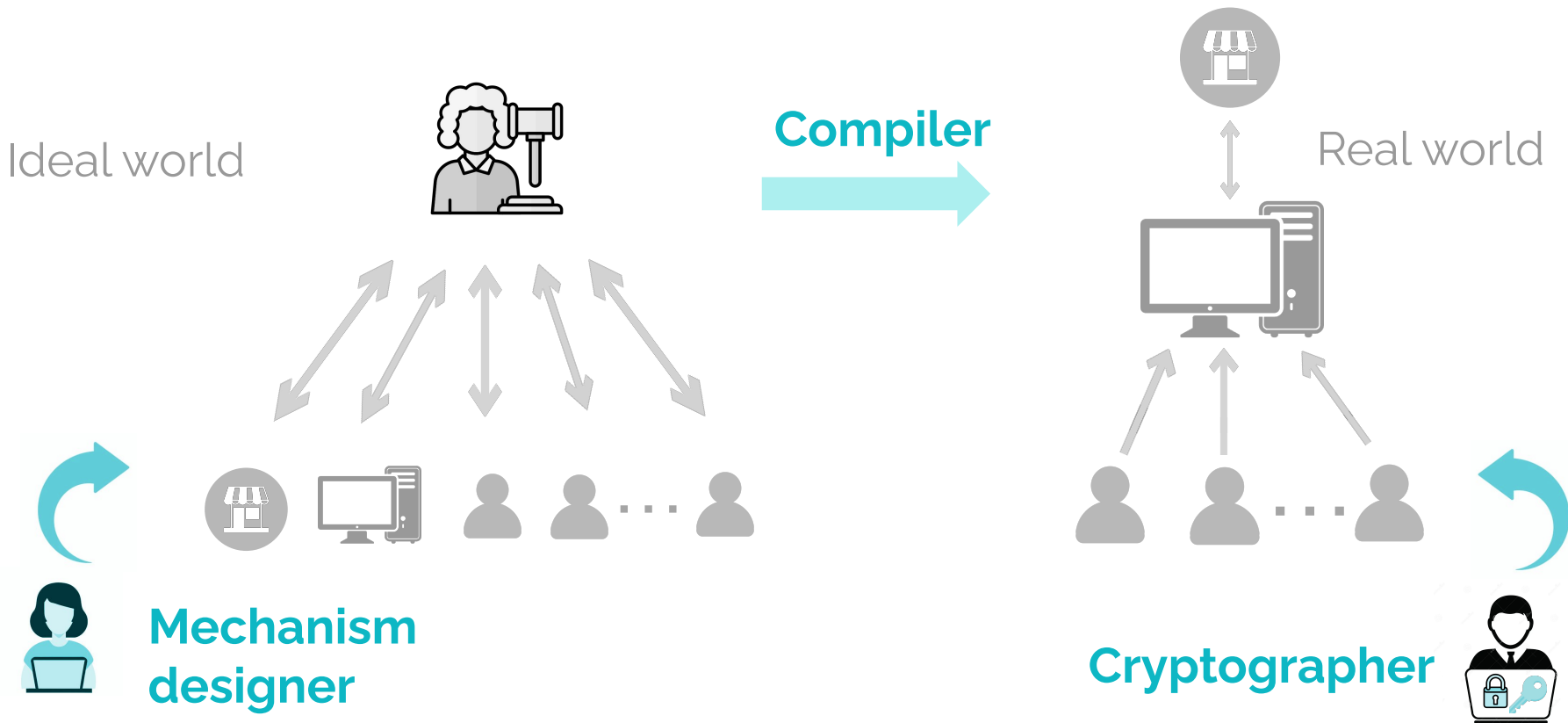


Generic MPC incurs n^2 cost!

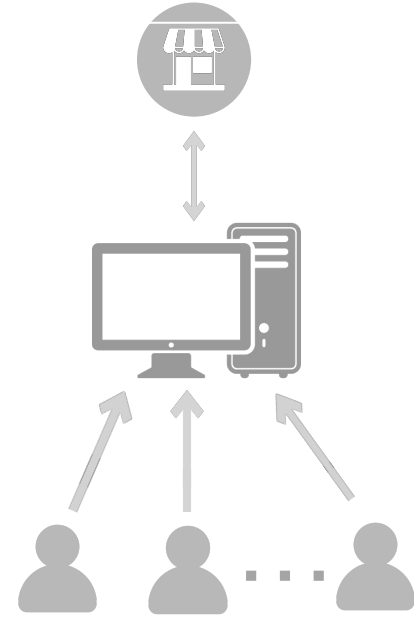
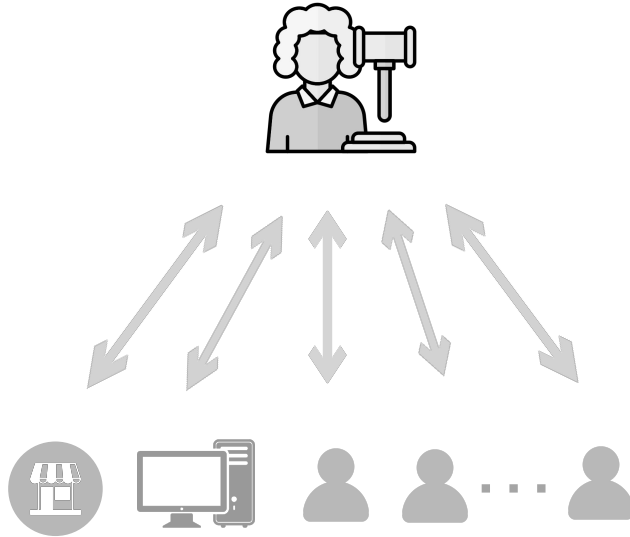
- Each player has a different output
- indistinguishability obfuscation
communication $\Rightarrow \sim O(n)$
compute: still n^2



Design paradigm of MPC



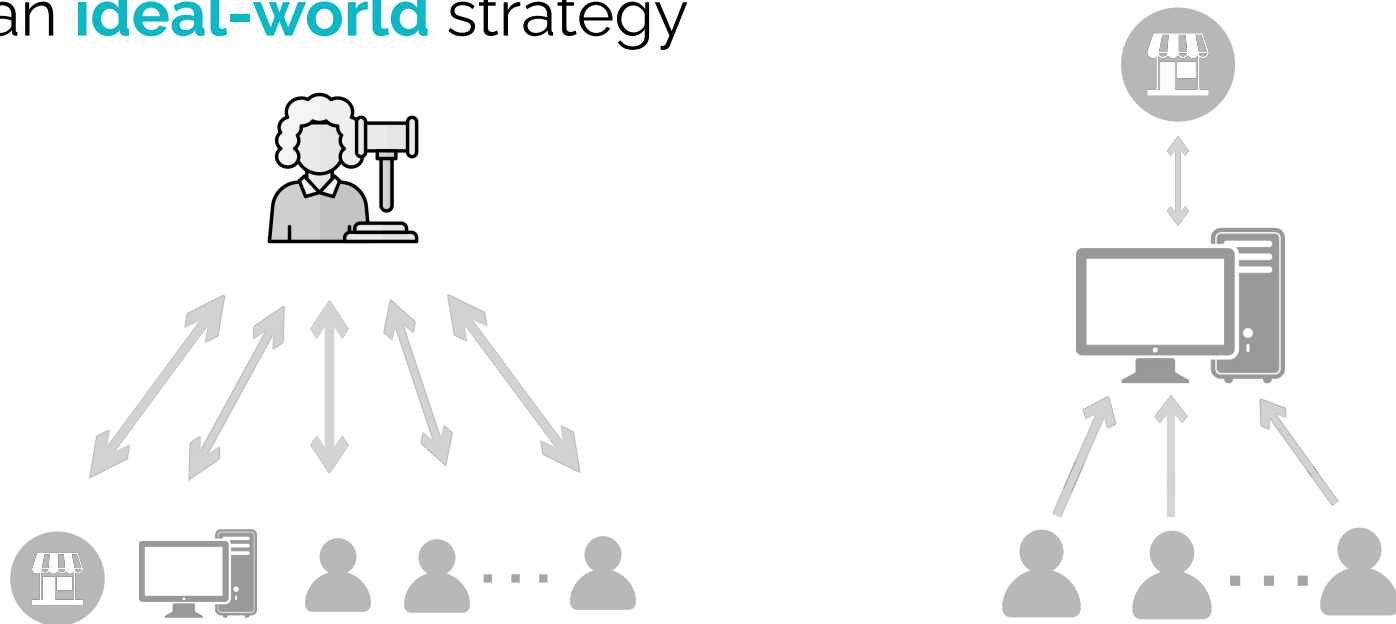
Can we improve the **efficiency**
but preserve the **design paradigm**?





Utility-dominated emulation:

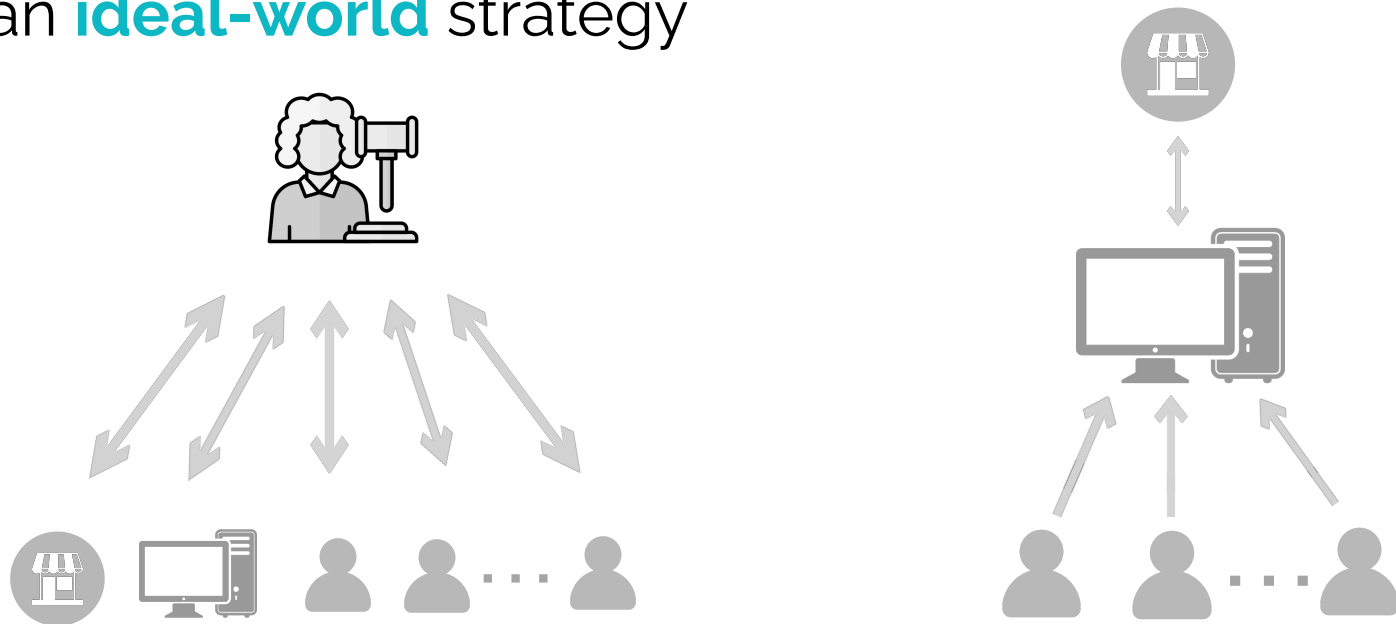
Any **real-world** strategy is **utility-dominated** by an **ideal-world** strategy





Utility-dominated emulation:

Any **real-world** strategy is **utility-dominated** by an **ideal-world** strategy



Thm: Ideal is IC + util-dominated emulation $\Rightarrow$ Real is IC



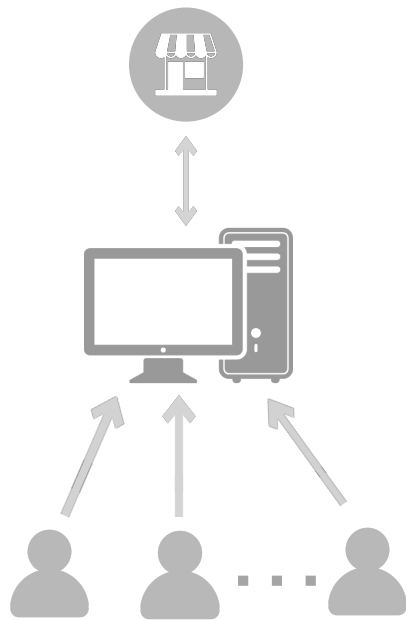
Our protocol

$\sim O(n)$ cost

$O(1)$ rounds

$O(1)$ broadcast

broadcast necessary due to
permissionless



See our paper for more

- More impossibilities & structural characterizations
- Efficient cryptography construction using ZK
- Computationally sound defn of “safe deviation”
- Proofs

<https://eprint.iacr.org/2025/019>

A photograph of an iceberg floating in the ocean. The tip of the iceberg is visible above the water surface, while the much larger, jagged mass of the iceberg is submerged below the surface. The water is a deep blue-green color. In the top left corner, there is a semi-transparent grey box containing the word 'Today' in white text, with a white arrow pointing from the text down towards the submerged part of the iceberg.

Today

Decentralized mechanism design:

a **goldmine** of open questions

- Biggest challenge for **blockchains**
- **Heuristic** protocols used in practice
- What's the **right game-theoretic notion**?
- **Crypto** meets mechanism design

Thank you !

elainershi@gmail.com



The protocol

1 Buyers send timed commitments of bids

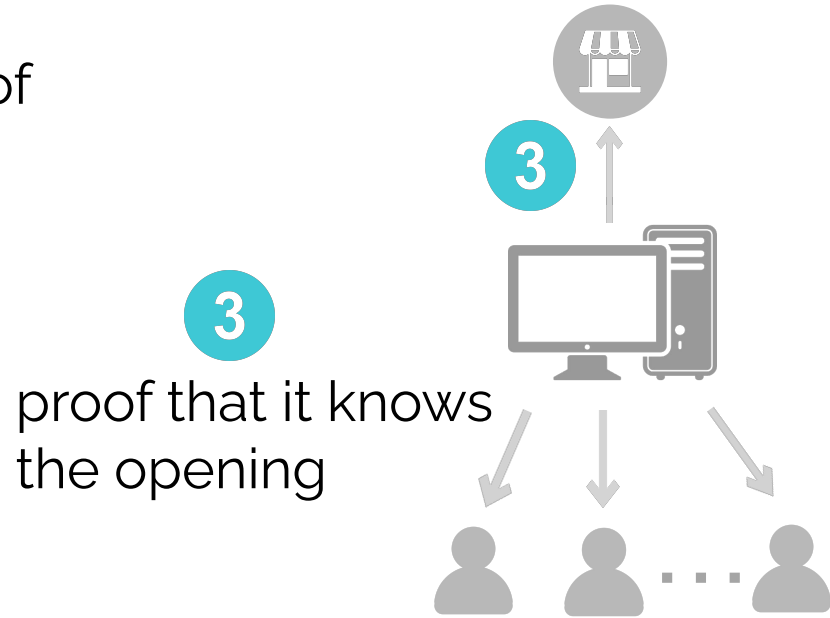
1 time-commit(bid)



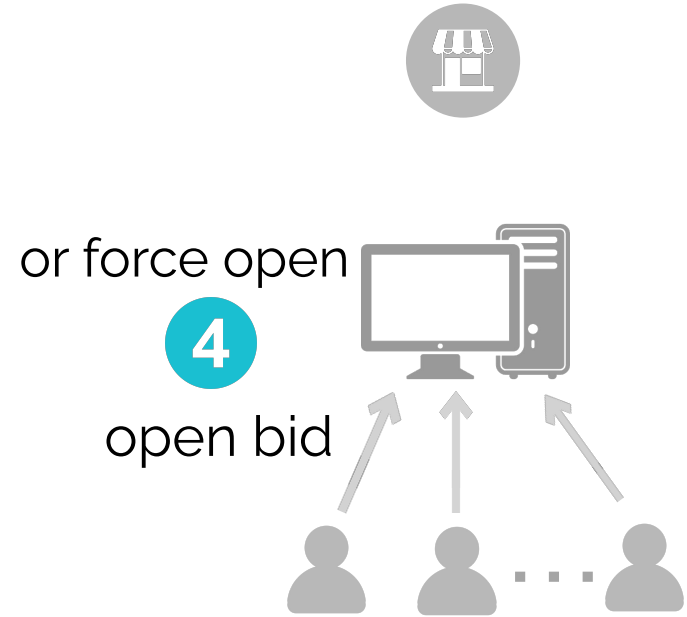
- 1 Buyers send timed commitments of bids
- 2 Platform broadcasts hash of commitments



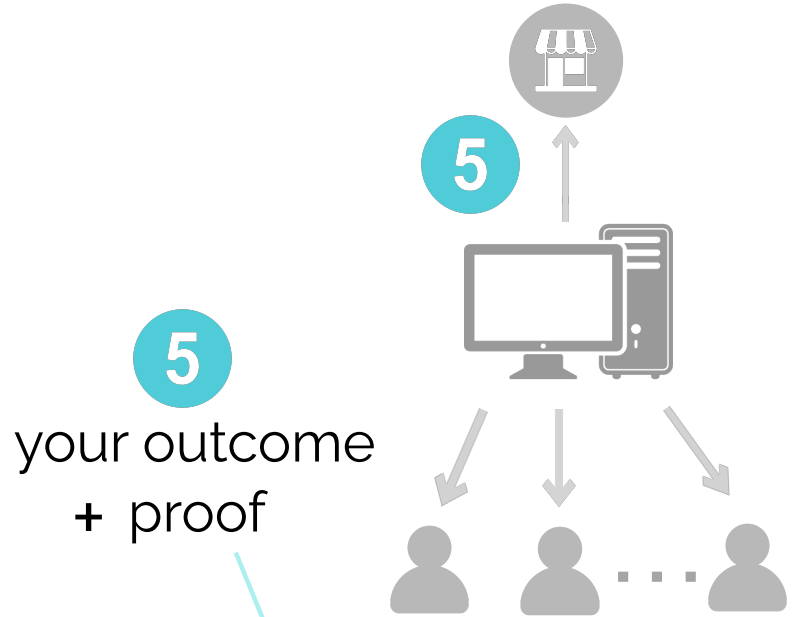
- 1 Buyers send timed commitments of bids
- 2 Platform broadcasts hash of commitments
- 3 Platform proves it knows opening of hash



- 1 Buyers send timed commitments of bids
- 2 Platform broadcasts hash of commitments
- 3 Platform proves it knows opening of hash
- 4 Open or force-open bids



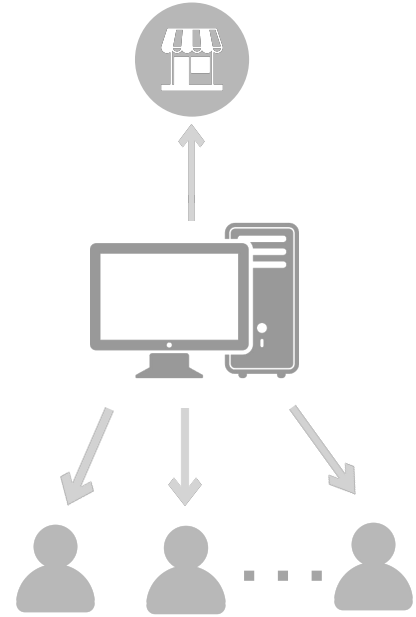
- 1 Buyers send timed commitments of bids
- 2 Platform broadcasts hash of commitments
- 3 Platform proves it knows opening of hash
- 4 Open or force-open bids
- 5 Platform sends everyone its outcome + proof



- outcome correct w.r.t. hash
- buyer's bid is included once

- 1 Buyers send timed commitments of bids
- 2 Platform broadcasts hash of commitments
- 3 Platform proves it knows opening of hash
- 4 Open or force-open bids
- 5 Platform sends everyone its outcome + proof

$O(1)$ rounds
 $O(1)$ bcast
 $\sim O(n)$ cost



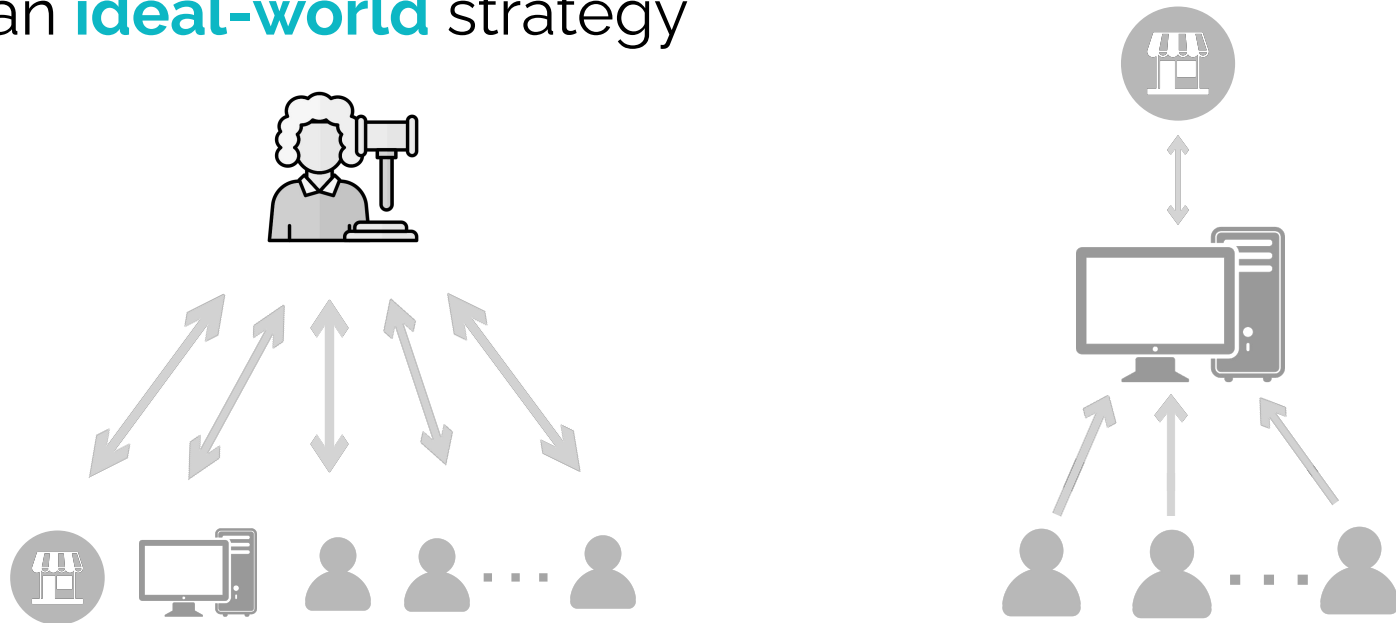
Thank you !

elainershi@gmail.com



Utility-dominated emulation:

Any **real-world** strategy is **utility-dominated** by an **ideal-world** strategy



$$\begin{aligned} \text{Strategic util in Real} &\leq \text{Strategic util in Ideal} \leq \text{Honest util in Ideal} \\ &= \text{Honest util in Real} \end{aligned}$$